

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 9: Dead Reckoning & Flight Computer Principles (CRP-5)

Syllabus Source: CAE Oxford (pp. 325-364) | Pass Target: $\geq 90\%$ AviationExam

1. DEAD RECKONING (DR) FUNDAMENTALS

Dead Reckoning is the navigation method of calculating current and future aircraft position from a previously known position using heading, airspeed, wind, and elapsed time:

- The Navigation Vector Triangle: Heading Vector (True Heading + TAS) + Wind Vector (Wind Direction + Wind Speed) = Track Vector (True Track + Groundspeed).
- The CRP-5 / E6B Slide Rule: Circular slide rule solving: 1. Time-Speed-Distance (Time = Dist / GS); 2. Fuel consumption (Burn = Time x Fuel Flow); 3. TAS from Calibrated Airspeed (CAS), Pressure Altitude, and Temperature (correcting for density error and compressibility).

2. TRUE AIRSPEED (TAS) VS INDICATED AIRSPEED (IAS)

- Rule of Thumb: TAS INCREASES by approximately 1.5% to 2.0% per 1,000 ft of altitude above sea level for a given Indicated Airspeed!
- Mach Number & Temperature: Mach = TAS / Speed of Sound (a). Speed of Sound depends SOLELY on Absolute Temperature (Kelvin): $a = 38.94 \times \sqrt{T_K}$ knots. In colder air, sound travels slower!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Wind Triangle & Flight Computer

Problem: Desired Track = 180° . TAS = 200 kt. Wind = $270^\circ/40$ kt. Step 1: Relative wind angle = $270^\circ - 180^\circ = 90^\circ$ (pure right crosswind). Step 2: Head/Tail component = $40 \times \cos(90^\circ) = 0$ kt. Step 3: Crosswind component = $40 \times \sin(90^\circ) = 40$ kt from right. Step 4: Wind Correction Angle (WCA): $\sin(WCA) = 40 / 200 = 0.20 \rightarrow WCA = 12^\circ$ to the right. Step 5: True Heading = $180^\circ + 12^\circ = 192^\circ$. Step 6: Groundspeed = $\sqrt{200^2 - 40^2} = 196$ kt!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Estimating TAS from IAS and Altitude

SCENARIO (Basic Navigation Rule of Thumb Drill): An aircraft climbs to FL 100 (10,000 ft): - Indicated Airspeed (IAS / CAS) = 150 knots - Outside Air Temperature is standard ISA ($+15^\circ\text{C} - 20^\circ\text{C} = -5^\circ\text{C}$) QUESTION: What is the approximate True Airspeed (TAS)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Recall the standard rule of thumb for TAS increase with altitude: - Rule: Add 2% to IAS per 1,000 ft of altitude! Step 2: Calculate the total percentage increase for 10,000 ft: - Altitude = 10,000 ft = $10 \times 1,000$ ft. - Total percentage increase = $10 \times 2\% = +20\%$! Step 3: Calculate the knots to add to IAS: - 20% of 150 knots = 0.20×150 kt = 30 knots. Step 4: Add the increase to the IAS: - TAS = IAS + Increase = 150 kt + 30 kt = 180 knots! FINAL ANSWER: True Airspeed (TAS) is approximately 180 knots (a 30 knot gain over indicated speed due to lower air density at 10,000 ft).

[!] EXAM TRAP (AVIATIONEXAM) - The Speed of Sound Depends Only on Temperature Trap

- The Speed of Sound DOES NOT depend on pressure or air density! - It depends SOLELY on absolute temperature: $a = 38.94 \times \sqrt{T_{\text{Kelvin}}}$! - Exam questions asking if sound speed changes with pressure are traps: the answer is NO, only temperature matters!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the primary physical reason why True Airspeed (TAS) is greater than Indicated Airspeed (IAS) at high altitude?**

- [A]: Engine develops more thrust
- [B]: Air density DECREASES with altitude, requiring the aircraft to travel faster through the thin air to produce the same dynamic pressure ($q = 0.5 \rho V^2$)
- [C]: Less gravity
- [D]: Instrument error

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The ASI measures dynamic impact pressure. At lower density, the aircraft must fly faster to register the same indicated pressure.

Q2: On which single atmospheric parameter does the local SPEED OF SOUND depend?

- [A]: Atmospheric pressure
- [B]: ABSOLUTE TEMPERATURE (in Kelvin)
- [C]: Air density only
- [D]: Wind speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In a perfect gas, sound speed is given by $a = \sqrt{\gamma \times R \times T}$. Only absolute temperature T affects sound velocity.

Q3: If an aircraft is cruising at FL 300 with an IAS of 240 kt in standard ISA conditions (-45°C), what is the approximate True Airspeed (TAS)?

- [A]: 300 kt
- [B]: Approx 384 kt ($240 \text{ kt} + 60\% = 384 \text{ kt}$)
- [C]: 240 kt
- [D]: 450 kt

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. $30,000 \text{ ft} \times 2\% / 1,000 \text{ ft} = +60\%$. $240 \times 1.60 = 384 \text{ knots TAS}$.